Quantifying Cohesion: Behavioral Clustering and Markov State Transitions Measure Team Structures that Lead to Goals

Introduction

In modern sports analytics, evaluating individual player performance is well-documented, but mathematically quantifying **team cohesion** remains a complex challenge. Here we ask: Can behavioral clustering and Markov state transitions capture the dynamic team structures that lead to goals? To answer this, we developed an inferential framework to analyze spatial and behavioral team dynamics. First, we defined the game's structure by applying K-Means clustering to a 10-dimensional player-description vector, categorizing players into discrete "active" or "inactive" behavioral states. Next, we measured the statistical impact of these states using t-tests across multiple time windows preceding an offensive event. This analysis demonstrated that active, cohesive team configurations are significantly different in the immediate lead-up to a goal compared to an unsuccessful shot. Finally, we mapped the offensive flow by designing a Markov Chain to model how teams transition between these behavioral states. By treating the game as a sequence of quantifiable behavioral states, this analysis provides actionable, statistical evidence of how synchronized movement generates high-danger scoring opportunities.

Defining Behavioral States

To define whether players were active or inactive, we first collected 10 metrics of activity. These were *Velocity* (smoothed with a 9-frame centered rolling mean, and capped at 38.28 ft/s), *Acceleration* (capped between -50ft/s and 16.4ft/s^2), *Absolute Acceleration* (smoothed with a 9-frame centered rolling mean), the derivative of acceleration that we labeled *Jerk* (smoothed with a 9-frame centered rolling mean), the *Closing Speed on Puck* (the negative derivative of distance-to-puck, so positive number means approaching the puck), *Distance to Team Centroid*, *Distance to Net* (minimum of distance to either net), *In-Crease* (binary, whether a player was within 8ft of the nearest net), *In-Slot* (binary, whether a player was within 20ft of the nearest net), *Distance to Puck*, and *Around-Puck* (binary, within 5ft of the puck). These metrics were calculated and then downsampled to 1 observation per 0.5 seconds (or 2Hz).

Once we had the metric vector for every player at every 0.5sec for every game, we ran unsupervised machine learning, K-Means clustering, on the vectors. We analyzed a range of clusters and used the elbow method to find the optimal number of clusters to keep. We analyzed the characteristics of each cluster to describe the behavioral state of said cluster and then used the definitions to build a binary “Activation” metric based on the metric values (Appendix Table 1.). “Activation” was defined as the player is either in a high-value location (slot/crease), holding the puck, or engaging in explosive kinetic movement.

Active Behavioral States[1]:

[1] Inactive Behavioral States: Cluster 2 – Transition; -1.45 ft/s *Closing Speed on Puck,* low Absolute *Acceleration* Cluster 4 – Perimeter Floater; 49.8 ft from *Distance to Puck,* 58.0 ft *Distance to Net*

- Cluster 0 – High Slot Presence; 14 ft to nearest net and 95% *In-Slot*
- Cluster 1 – High Exertion; High *Absolute Acceleration* and *Jerk*
- Cluster 3 – Puck Possession; 4 ft *Distance to Puck* and 82% *Around-Puck*
- Cluster 5 – Deep Crease; 7 ft *Distance to Net*, 87% *In-Crease*

Using only this cluster identification system, we can describe the player composition of teams and use it as another comparison metric between teams (Appendix Table 2.). Team G leads the league with 52% Total Activation, which is driven almost entirely by their league-leading 27% in the High Exertion state. This mathematically profiles them as a heavy, aggressive, high-energy team that thrives on forechecking and physical engagement. On the opposite end of the spectrum, Team B sits at just 36% Total Activation. They have the lowest engagement in the High Exertion state (13%) and the highest percentages in both inactive states (C2 at 41%, C4 at 23%). This profiles a team that plays a passive, perimeter-heavy game, perhaps relying on a neutral-zone trap or struggling to establish sustained offensive pressure.

We also see stability in the purely spatial clusters across all teams. Almost every team finds opportunities to have players spend roughly 14-15% of their time in the High Slot state (C0) and 3-4% in the Crease state (C5). This indicates that the differentiating factor between a highly active team and a passive team isn't necessarily their static formation, but their willingness to engage in the High Exertion states (C1) and maintain Puck Possession (C3), as well as how they transition between states.

Behavioral States in the Buildup to Goals

The clustering method introduced above is a new way to characterize how a team plays, and how teams differ from one another. Having established descriptive team profiles, we next tested whether these behavioral states offer predictive power over performance. We applied t-tests using two complementary approaches: peak concentration (max) across 20-, 10-, and 5-second windows, and sustained presence (mean) across tighter 2- and 1-second windows. Generalized team activation showed no statistically significant differences between goals and unsuccessful shots across any wide window (Figure 1, Panels A, B). Instead, conversion was tied to specific structural behavioral states operating on two distinct timescales. Across all three wide windows, goals were preceded by a significantly higher peak number of attacking players in the High-Slot state (C0, $p < 0.05$, Figure 1 Panel C), meaning that at some moment during the buildup, goal-scoring sequences achieved a greater maximum slot concentration than unsuccessful shots. In the final 1-2 seconds, a second mechanism emerges: goals are preceded by significantly higher sustained Crease state (C5, $p = 0.008$ at 1s, Figure 1 Panel D) and higher overall activation ($p = 0.025$ at 1s). The slot creates the shooting geometry; the crease creates the screen, the deflection, and the rebound. This two-phase pattern: set up the slot, then crash the crease, suggests that goals are not a product of generalized effort but of teams executing a specific structural sequence.

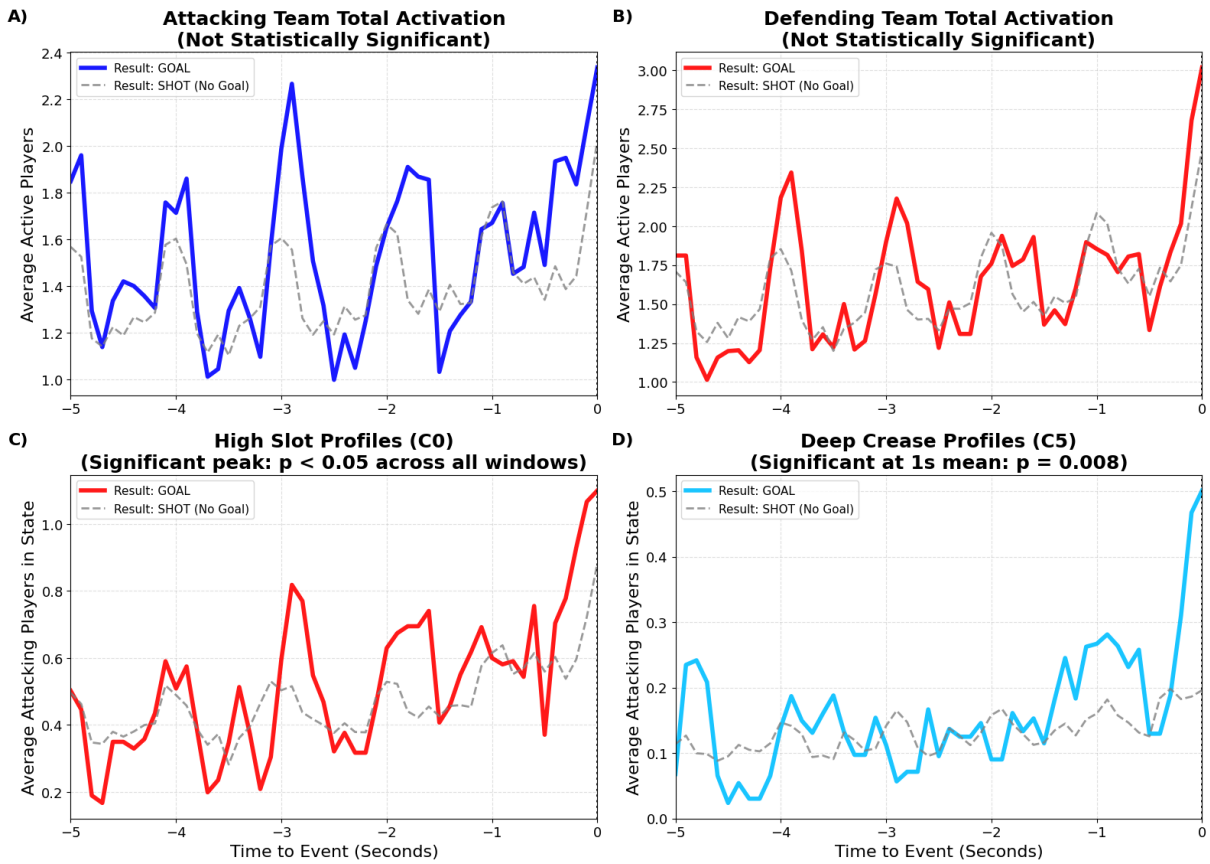

Figure 1. General Effort vs. Structural Buildup: 5-Second Lead-Up to Event. Panels A and B show total activation for attacking and defending teams, respectively. Neither shows a statistically significant difference between goals and shots across wide time windows. Panels C and D isolate specific behavioral states. Peak High Slot (C0) is significantly higher before goals across all three windows (20s, 10s, 5s; $p < 0.05$), indicating that goal-scoring sequences achieve a greater maximum slot presence at some point during the buildup. In the final 1–2 seconds, sustained crease presence (C5) emerges as a secondary differentiator ($p = 0.008$ at 1s).

Modeling Transitions

To model game flow, we treated the offensive zone as a discrete-time Markov Chain. By chronologically ordering every player's tracking data and calculating the state transitions over a $\Delta t = 0.5$ second interval, we derived the exact transition probabilities $P(S_{t+1} = j \mid S_t = i)$ between all behavioral states. This transition matrix allows us to measure two critical components of team structure: *State Stickiness* (the probability a player remains in their current behavioral state) and *Tactical Pathways* (the probability of transitioning into a new, "higher value" behavioral state).

The global transition matrix (Figure 2) reveals that offensive hockey is defined by long periods of structural stability punctuated by micro-moments of chaos, as seen by the high probabilities of staying in the same state. Structural states, like gliding in transition or anchoring in the high slot, show over 65% stickiness, meaning players who enter these behavioral states tend to hold them. The High Exertion state is highly unstable (39% stickiness): 38% of the time players disengage

into a glide, 12% are pushed to the perimeter, and only 4% emerge with clean possession. The matrix also captures specific spatial behaviors like the "Crease Pop-Out," where nearly 33% of players in the deep crease transition directly into the high slot in the next half-second, modeling the behavior of a net-front player backing into soft ice.

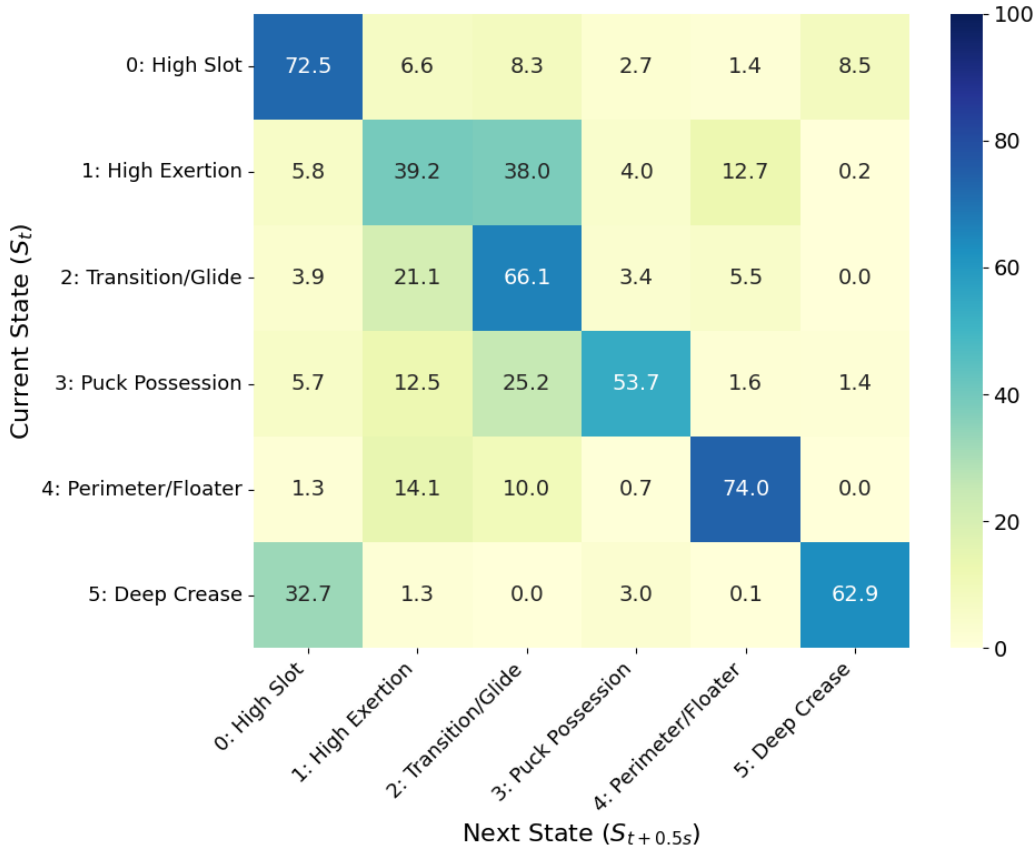

Figure 2. Global Markov Chain: State Transition Probabilities (%). Each cell shows the probability of transitioning between behavioral states over a 0.5-second interval for all teams. Diagonal values represent *state stickiness*. The High Slot (72.5%) and Perimeter/Floater (74.0%) values on the diagonal show high persistence, while High Exertion (39.2%) is the most volatile. Notable off-diagonal transitions include the "Crease Pop-Out" (C5→C0, 32.7%) and the low rate of clean possession emerging from states of High Exertion (C1→C3, 4.0%).

When isolated by team, distinct tactical identities emerge (Figure 3). Team L defines the 'Structural' approach, leading the league in transitioning from the High Slot to the Puck Carrier behavioral state, *Slot Activation* (C0→C3: 3.83%) while maintaining one of the lowest dependencies on physical puck-recoveries. This suggests a system where superior spacing and passing allow players to receive the puck in dangerous areas.

In contrast**,** Team B represents a 'Passive-Exertion' profile. Despite having the league's lowest overall engagement in physical high exertion states (13%, Appendix Table 2), they possess the highest recovery dependency; they lead the league in transitioning from High Exertion to the Puck Carrier state, *Puck Recovery* (C1→C3: 5.8%). This indicates that while Team B spends most of their time outside of High Exertion behavioral states, they lack the structural geometry to

move the puck effectively, forcing them to rely on isolated bursts of individual physical effort to 'rescue' possession and move it toward the net.

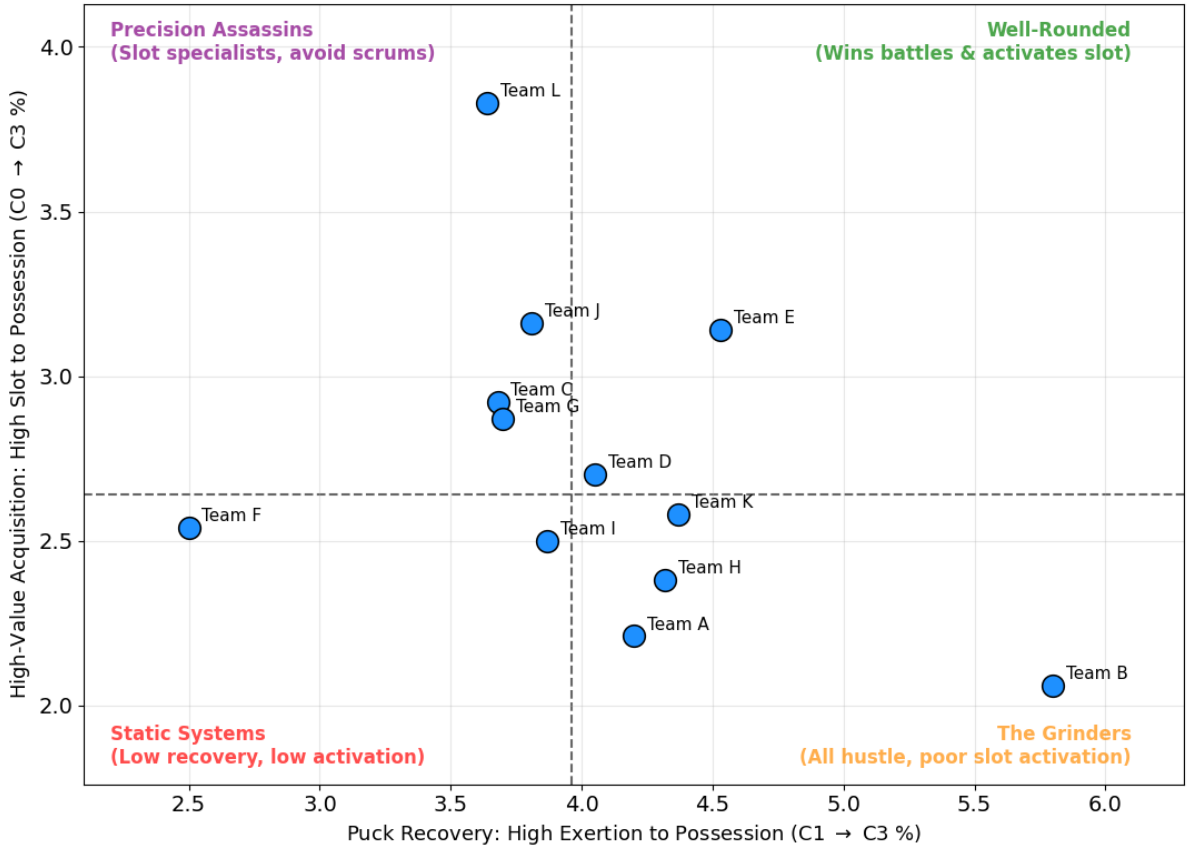

Figure 3. Teams are plotted by *Puck Recovery* (C1→C3, x-axis) and *Slot Activation* (C0→C3, y-axis). The four quadrants reveal distinct profiles, from structure-dependent systems like Team L (high *Slot Activation*, low *Puck Recovery*) to exertion-dependent systems like Team B (high *Puck Recovery*, low *Slot Activation*).

Applications

The transition probabilities of game flow have direct tactical utility. For line construction, pairing players with high *Puck Recovery* (C1→C3) with high *Slot Activation* (C2→C0) players mathematically ensures the player winning the puck in the corner has a teammate sliding into the shooting pocket. Consider Team B, with strong *Puck Recovery* but weak *Slot Activation*. Their transition data would suggest pairing a forechecking wing who wins board battles at high rates with a center who consistently reads the play and drifts from transition into the high slot, increasing their *Slot Activation* probabilities. The specific players could be identified by examining individual players' time distributions across clusters. The Markov probabilities make this quantifiable rather than intuitive. For opponent scouting, defensive game plans should target an opponent's most dominant pathway. Against structure-dependent teams like Team L, which move the puck primarily through the High Slot rather than engage in physical battles, opponents should look to clog passing lanes to the High Slot rather than over-committing to board battles.

Limitations

This report introduces a new framework for quantifying behavioral states, but several limitations should be noted. The analysis did not distinguish between game situations. Power play, even strength, and overtime data were all pooled together, and these specific events may produce different clustering results. The optical tracking system also introduced data quality challenges: tracking glitches and missing data were prevalent, and while preprocessing addressed the most obvious errors, undetected gaps may distort transition probabilities in localized segments. Beyond data quality, the tracking data was down-sampled to 2Hz, which filters optical jitter but may obscure rapid micro-transitions like touch-passes occurring between half-second snapshots. K-Means clustering introduces subjectivity in how cluster behaviors are interpreted, with our descriptions reflecting dominant tendencies, not absolute labels. The 10-dimensional vector, though robust, does not capture player orientation or defensive pressure, and the spatial metrics (*Crease*, *Slot*) use simple distance thresholds rather than precise rink geometry. Finally, the same cluster definitions were applied to both attacking and defending behavioral states, which may mask important asymmetries in how offensive and defensive behaviors manifest.

Future Directions and Conclusions

Future work should integrate play-by-play passing data to distinguish coordinated behavioral state transitions from chaotic turnovers and reverse the framework to identify which defensive formations most effectively disrupt teams' dominant behavioral states. Re-running the pipeline at higher frame rates (5–10Hz) would capture fluid micro-events currently lost between snapshots. Beyond team-level analysis, pairwise player transition states could quantify on-ice chemistry between specific linemates, and tracking individual players' time distributions across clusters could identify which players most effectively coordinate attacks and deliver pucks to high-danger locations.

This project demonstrates that team cohesion can be mathematically quantified. By clustering players into discrete behavioral states, we showed that team compositions reveal distinct tactics across teams and games. By applying significance testing to the time windows preceding shots, we found that goals were not driven by generalized effort but by a two-phase structural sequence: peak concentration in the High Slot during the buildup, followed by sustained Crease presence in the final 1–2 seconds before the shot. By modeling these states as a Markov Chain, we mapped the specific transition pathways that separate structured offensive systems from exertion-dependent ones. Together, these methods offer coaches and analysts a framework for evaluating team structure with the same statistical rigor traditionally reserved for individual player performance. As optical tracking data becomes standard across leagues, frameworks like this will shift tactical evaluation from subjective film review toward reproducible, data-driven structural analysis of team cohesion and strategy.

GitHub: https://github.com/nmorok/Big-Data-Cup-2026

Appendix:

Cluster	Velocity (ft/s)	Absolute Acceleration (ft/s2)	Jerk (ft/s3)	Distance to Puck (ft)	Closing Speed on Puck (ft/s)	Distance to teammate Centroid (ft)	Distance to nearest net (ft)	% in Crease	% in Slot	% around puck	Description
0	11.89	11.6	34.96	26.89	0.12	15.58	14.08	2%	95%	1%	ACTIVE - High slot presence. The shooter's pocket between the hash marks.
1	17.89	18.06	55.84	27.72	-1.1	18.09	48.09	0%	4%	1%	ACTIVE - Explosive battles/scrums. High acceleration and jerk, violent direction changes, or board play.
2	15.83	10.16	30.02	23.76	-1.45	14.5	49.76	0%	1%	1%	INACTIVE - Standard transition/gliding. Positional tracking, or skating up/down the ice away from the play.
3	15.81	12.85	38.95	4.04	0.39	18.25	45.14	1%	16%	82%	ACTIVE - Puck possession. The puck carrier or the player immediately engaged with the puck.
4	14.88	10.61	31.88	49.82	-1.32	29.45	57.96	0%	0%	0%	INACTIVE - Perimeter floater. Farthest from puck/centroid; coasting on the weak side, or line changes.
5	10.73	12.03	36.39	23.6	-1.34	15.69	6.72	87%	100%	6%	ACTIVE - Deep crease. Closest to net; screening the goalie, jamming rebounds, or net-front defense.

Table 1. Table showing the mean values of all of the behavioral metrics from the positional vectors. The cluster descriptions were written using best available data and do not represent the only times that the players are labeled with that cluster, only a representation of common behavior. Cells highlighted in yellow show strong differentiators between other clusters.

Team	Total Activation	C1 (High Exertion)	C2 (Transition)	C4 (Perimeter)	C0 (High Slot)	C5 (Crease)
Team G	52%	27%	34%	14%	16%	4%
Team L	49%	27%	33%	18%	13%	4%
League Avg	*44.5%*	*20.8%*	*36.3%*	*19.1%*	*14.3%*	*3.4%*
Team A	42%	18%	38%	21%	15%	3%
Team B	36%	13%	41%	23%	15%	3%

Table 2. Select team composition defined by the percentage of total player-time spent in each behavioral state. "Total Activation %" aggregates the time spent in the four high-impact states (Clusters 0, 1, 3, and 5). This macro-level view mathematically quantifies team playstyles, contrasting highly aggressive, high exertion teams (e.g., Team G) with passive, perimeter-oriented teams (e.g., Team B).